

MonsoonIQ — Verification Report

Regime-aware post-processing of NWP rainfall · held-out period [2025] · 121 days · 6413 district-days · generated 2026-09-25 13:58 UTC

DATA PROVENANCE: REAL_OBSERVED_IMD_NWP — Observed IMD 0.25 deg gridded rainfall aggregated to Census-2011 district polygons; raw forecasts are archived operational NWP runs. Skill figures on this archive are genuine out-of-sample verification. All skill figures below are measured against this archive and are **not** evidence of operational skill.

1 · Scorecard (held-out period)

Quantity	Raw NWP	Regime-agnostic	Regime-aware	Change vs raw	Change vs baseline
RMSE (mm/day)	13.950	13.169	13.333	-0.617	+0.164
Heavy-rain CSI (≥64.5 mm)	0.051	0.104	0.085	+0.0332 (significant)	-0.0194 (significant)
Heavy-rain ETS	0.044	0.090	0.073	+0.029	-0.016
Very heavy CSI (≥115.6 mm)	0.018	0.012	0.059	+0.041	+0.047

The regime-agnostic column is a gradient-boosted model fitted on the same predictors without regime information. It is the baseline that decides whether regime conditioning earns its complexity; the raw model alone does not.

1b · District warning product vs observed district maximum

Lead	Threshold	Events	Raw POD	Raw CSI	Warning POD	Warning FAR	Warning CSI [95% CI]	Freq. bias
Day 1	≥64.5	626	0.053	0.051	0.570	0.566	0.327 [0.290, 0.366]	1.31
Day 1	≥115.6	167	0.018	0.018	0.389	0.693	0.207 [0.148, 0.269]	1.27
Day 2	≥64.5	626	0.050	0.048	0.554	0.607	0.299 [0.258, 0.340]	1.41
Day 2	≥115.6	167	0.012	0.012	0.234	0.715	0.147 [0.093, 0.202]	0.82
Day 3	≥64.5	626	0.053	0.051	0.460	0.562	0.289 [0.245, 0.332]	1.05
Day 3	≥115.6	167	0.000	0.000	0.234	0.719	0.146 [0.101, 0.191]	0.83
Day 4	≥64.5	626	0.045	0.043	0.444	0.580	0.275 [0.231, 0.321]	1.06
Day 4	≥115.6	167	0.000	0.000	0.317	0.730	0.171 [0.118, 0.225]	1.17
Day 5	≥64.5	626	0.029	0.027	0.428	0.564	0.276 [0.231, 0.319]	0.98
Day 5	≥115.6	167	0.000	0.000	0.204	0.682	0.142 [0.086, 0.199]	0.64

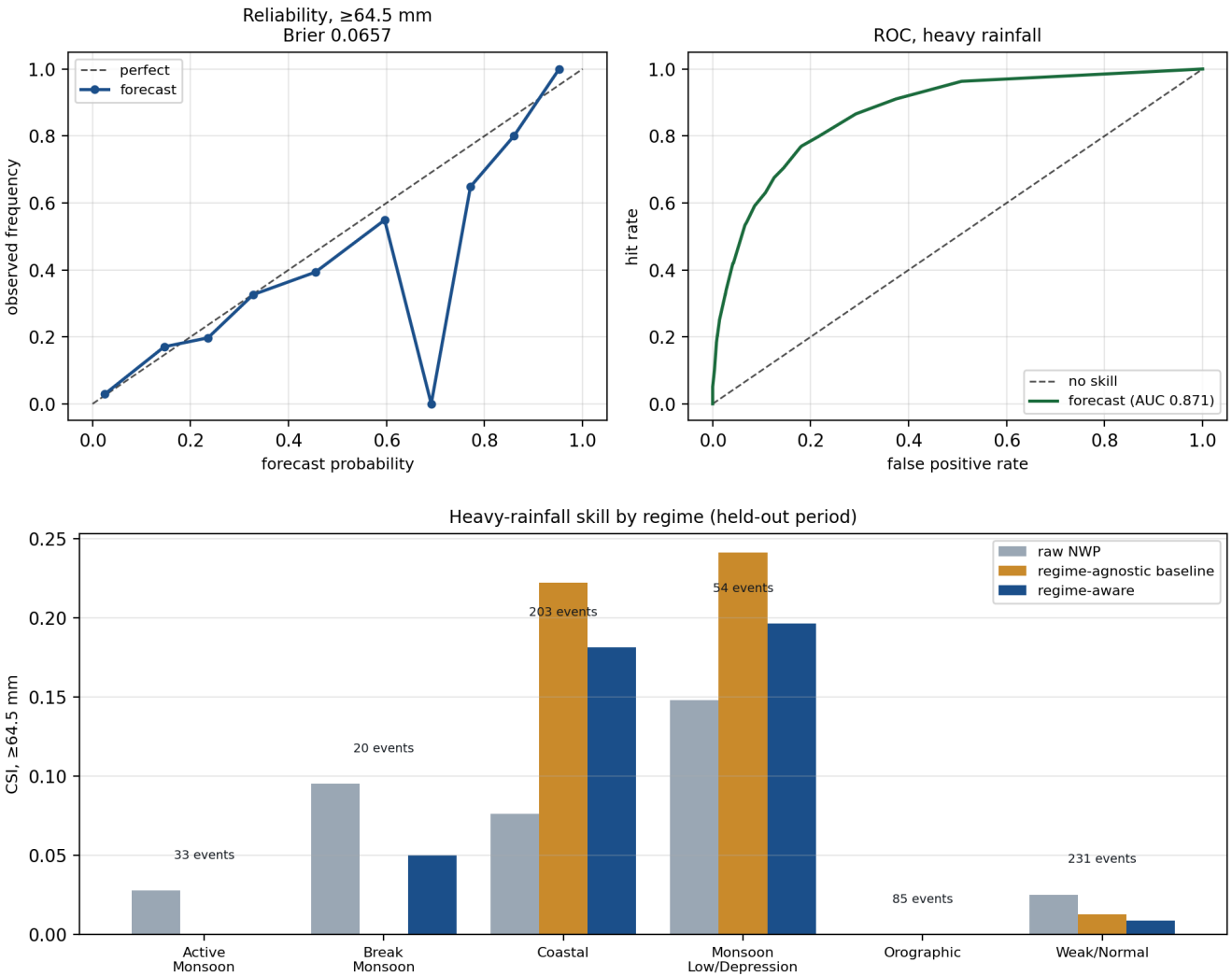
IMD heavy-rain warnings concern the heaviest rain in a district, so the product (calibrated exceedance probabilities, P90 and corrected amount, combined by the same rule the console and bulletin use) is scored against the district maximum of the 0.25° cells. The raw model is read the same way (district forecast ≥ threshold).

2 · Heavy and very heavy rainfall, by regime

Regime	Days	Events (≥64.5)	≥115.6	Reliability	Raw CSI	Regime-agnostic CSI	Regime-aware CSI	Δ vs raw [95% CI]
Active Monsoon	351	33	6	reliable	0.028	0.000	0.000	-0.028 [-0.092, +0.000]
Break Monsoon	255	20	3	indicative	0.095	0.000	0.050	-0.045 [-0.185, +0.006]
Coastal	1035	203	56	reliable	0.076	0.222	0.181	+0.105 [+0.063, +0.151]
Monsoon Low/Depression	199	54	27	reliable	0.148	0.241	0.196	+0.048 [-0.025, +0.121]
Orographic	707	85	29	reliable	0.000	0.000	0.000	+0.000 [+0.000, +0.000]
Weak/Normal	3866	231	46	reliable	0.025	0.013	0.009	-0.017 [-0.034, -0.003]

Strata with fewer than 10 events are marked **insufficient**: their CSI is printed for completeness only and is not a scored result. Intervals are day-block bootstraps — whole calendar days are resampled, because 53 districts on one day are a single weather system.

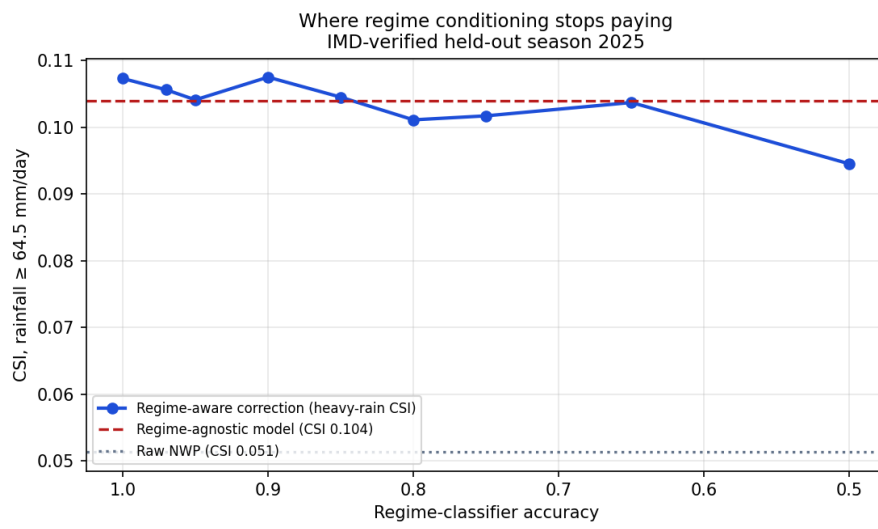
3 · Reliability, ROC and regime breakdown



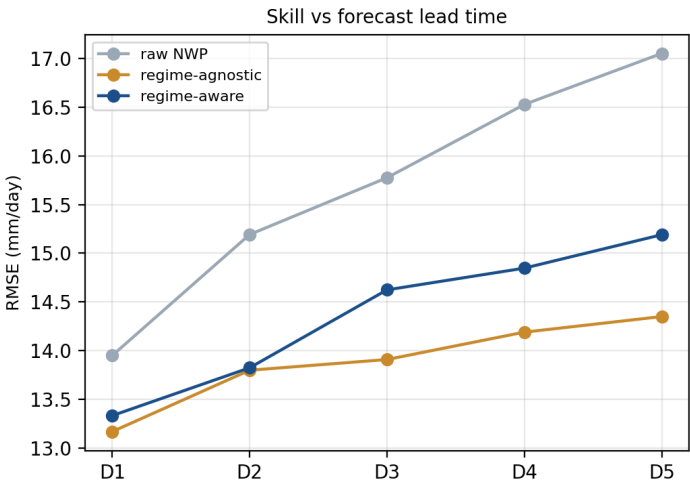
4 · What regime awareness is worth

Experiment	Result
Correction alone, uniform regime weights (CSI)	0.0574
Correction + classifier regime weights (CSI)	0.0845
Regime-agnostic model, same predictors (CSI)	0.1039
Gain attributed to regime awareness	-0.0194 [-0.0383, -0.0024] — significant
Classifier accuracy at which the gain vanishes	0.85

Uniform regime weights reproduce a 'correction with no regime information' system; the gap between it and the classifier blend is the actual value of regime awareness. If that gap is inside the interval, the regime layer is not yet earning its complexity and the honest fix is more training seasons or a better regime classifier - not a stronger claim.



5 · Lead time and spatial scale



6 • Claim check

Claim	Verdict	Evidence
Post-processing reduces RMSE relative to the raw model	NOT SUPPORTED	Δ RMSE -0.607 mm/day, 95% CI [-1.262, +0.022]
Post-processing improves the heavy-rainfall categorical scores relative to the raw model	NOT SUPPORTED	Δ CSI +0.0332, 95% CI [+0.0100, +0.0543]
Regime conditioning beats a regime-agnostic model fitted to the same predictors (heavy rainfall categorical)	NOT SUPPORTED	Δ RMSE -0.0145 mm/day, 95% CI [-0.0389, +0.0024]
Regime conditioning beats a regime-agnostic model on RMSE	NOT SUPPORTED	Δ RMSE +0.167 mm/day, 95% CI [-0.820, +1.157]

Intervals resample whole calendar days. Comparisons use archived operational forecasts against IMD gridded observations on seasons never seen in training.

7 • Reproducibility

python src/train.py → python src/evaluate.py → pytest tests/ → uvicorn src.api.main:app
Model manifest: artifacts/models/training_manifest.json (Gradient boosting (regime-agnostic), MonsoonIQ (regime-aware), Quantile mapping (regime-agnostic), Raw NWP)
Every figure and number in this report is regenerated from the raw archive by the commands above; no value is hand-entered.